5.4 Summary

- The multi-armed bandit problem consists of a set of actions that may be performed (arms of slot machines that may be pulled), each of which returns a reward according to its specific reward rate. The goal is to maximize the sum of reward (cumulative reward) we receive given a number of iterations.
- A multi-armed bandit policy selects which action to take next based on past data. A good policy needs to balance between under-explored actions and high-performing actions.
- Unlike multi-armed bandit where there are a finite set of sections, there are infinitely many actions that we can take in Bayesian optimization.
- The goal of Bayesian optimization is to maximize the largest observed reward, which is often called the simple reward.
- The rewards in Bayesian optimization are correlated: similar actions yield similar rewards. This is not necessarily true in multi-armed bandit.
- The Upper Confidence Bound policy uses an optimistic estimate of the quantity of interest in order to make decisions. This "optimism in the face of uncertainty" heuristic can balance between exploration and exploitation with a tradeoff parameter that we, the users, can set.
- The smaller the tradeoff parameter of the Upper Confidence Bound policy is, the more exploitative the policy becomes, tending to stay within regions that are known to give high rewards. The larger the tradeoff parameter is, the more explorative the policy becomes, tending to query regions far away from observed data.
- The Thompson sampling policy draws a sample from a probabilistic model of the quantity of interest and uses this sample to make decisions.
- The random nature of Thompson sampling allows the policy to explore/exploit the search space appropriately: both regions of high uncertainty and those of high predictive mean are likely to be chosen by Thompson sampling.
- For computational reasons, more care is needed when implementing Thompson sampling: we first generate a set of points to cover the search space evenly, and then a sample from the Gaussian process posterior is drawn for these points.
- To cover a space evenly, we can use a Sobol sequence to generate the points within the unit cube and scale them to the targeted space.

5.5 Exercise

There are two exercises in this chapter:

1. The first explores a potential method to set the tradeoff parameter for the Upper Confidence Bound policy that considers how far along we are in optimization.
2. The second applies the two policies we have learned in this chapter to the hyperparameter tuning problem seen in previous chapters.

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